

# Endovascular Device for Rapid Blood Clot Removal

Zachary Huang, Nathan Ng, Michael Ramirez, Colleen Wang, Tony Wang

Department of Mechanical and Aerospace Engineering, University of California San Diego

Sponsored by Dr. Elsie Ross and Kayla Tracy

## Background

- > 800,000 individuals in the U.S. need hemodialysis due to chronic kidney disease [1]
- Requires **arteriovenous (AV) graft or AV fistula**

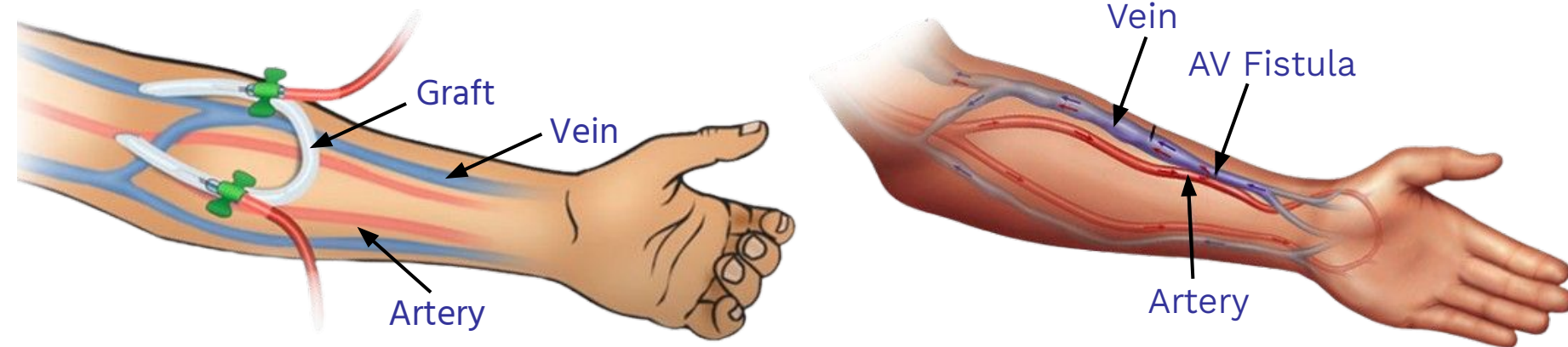


Fig. 1: Arteriovenous (AV) graft and fistula

- Prone to clotting**, which blocks hemodialysis access and leads to complications [2]
- Critical need for a mechanical thrombectomy **device that effectively and safely removes clots** in AV grafts/fistulas

## Existing Solutions

### Argon CLEANER™

- Difficult to maneuver
- Single sinusoidal wire
- 79% effective clot removal**
- > 1% tip breakage**



Fig. 2: Cleaner Device

### Arrow-Trerotola™ PTD(R)

- Easy to maneuver
- Expandable wire basket
- 83% effective clot removal**
- 21% tip breakage (recalled)** [3]

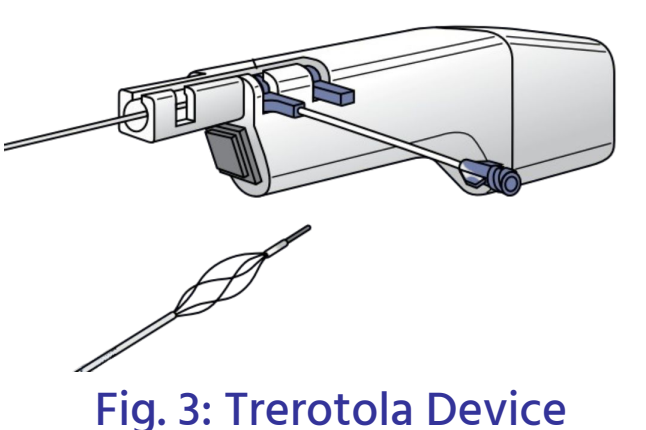


Fig. 3: Trerotola Device

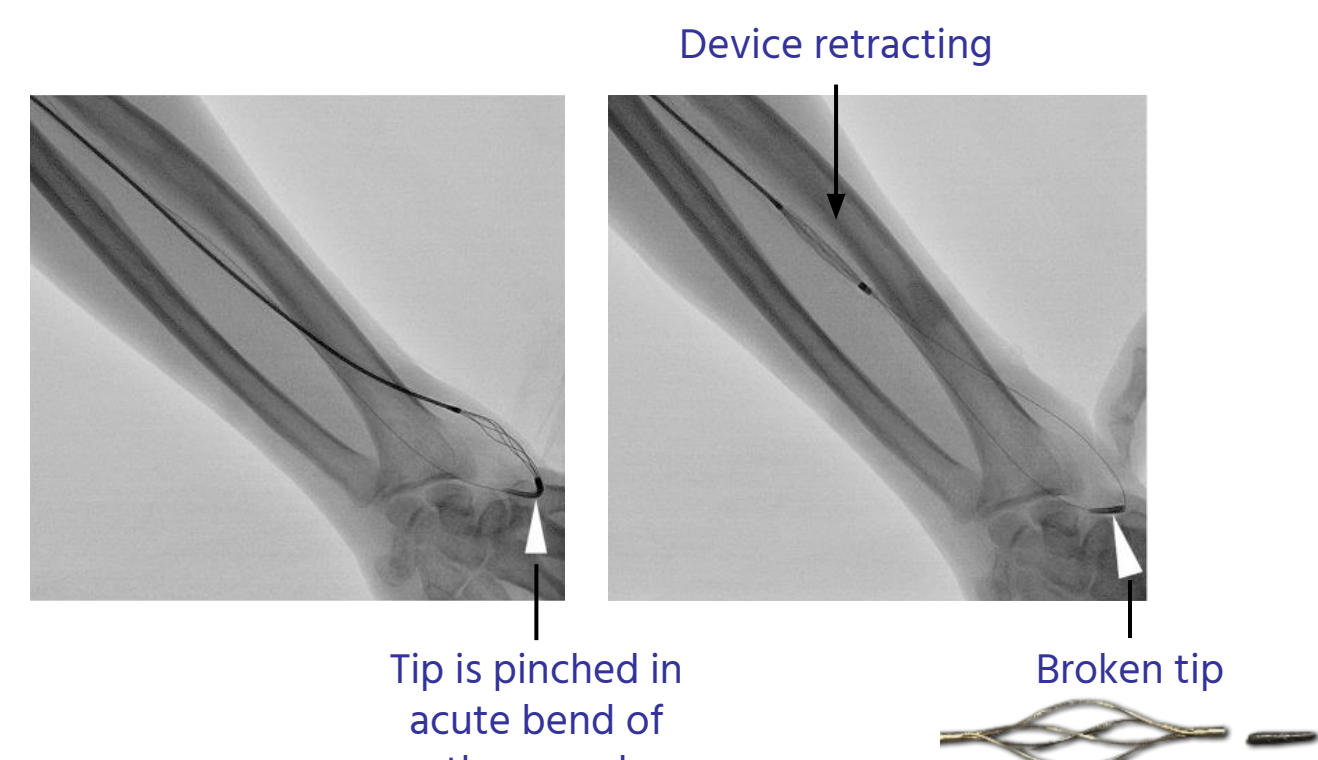


Fig. 4: Trerotola Tip Breakage Case

### 2024 Team's Ball-Bearing Prototype

- Easy to maneuver
- Expandable wire basket
- Innovative non-driven tip**
- 1.5x too large**

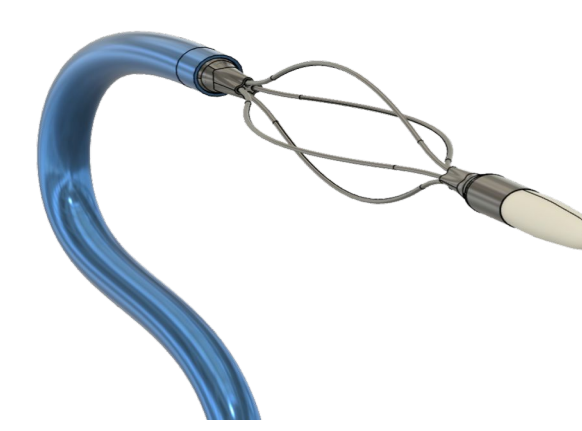


Fig. 5: Ball-Bearing Prototype

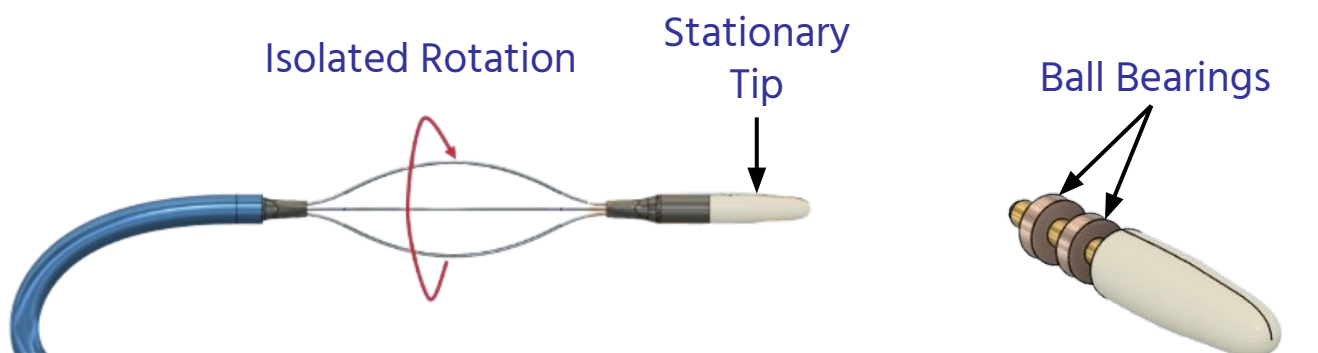


Fig. 6: Non-driven tip achieved with ball bearings

## Project Objectives

- Design a **non-driven tip** to fit within 2mm standard for endovascular procedures
- Custom drive unit** with variable speed control
- Enhance the existing testbed and establish **standardized clot removal efficiency measurement**

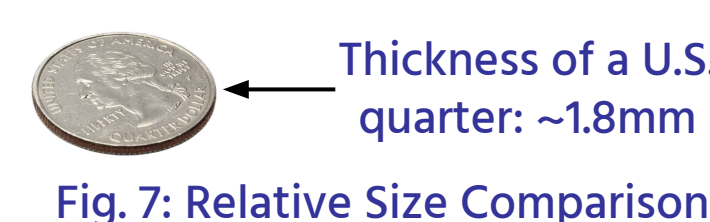


Fig. 7: Relative Size Comparison

## Team's Microbushing Design Prototype

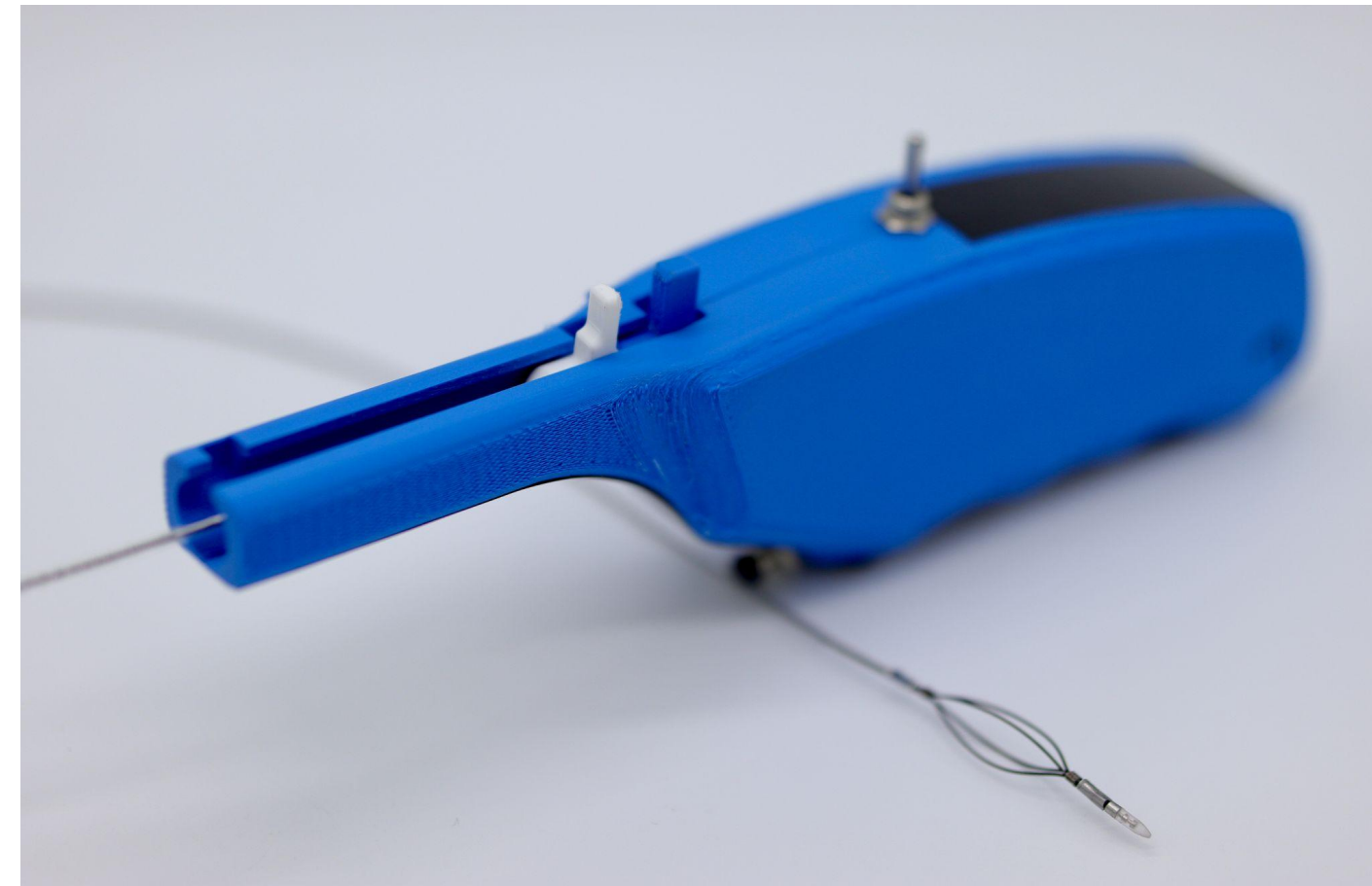


Fig. 8: Photo of Team's Prototype

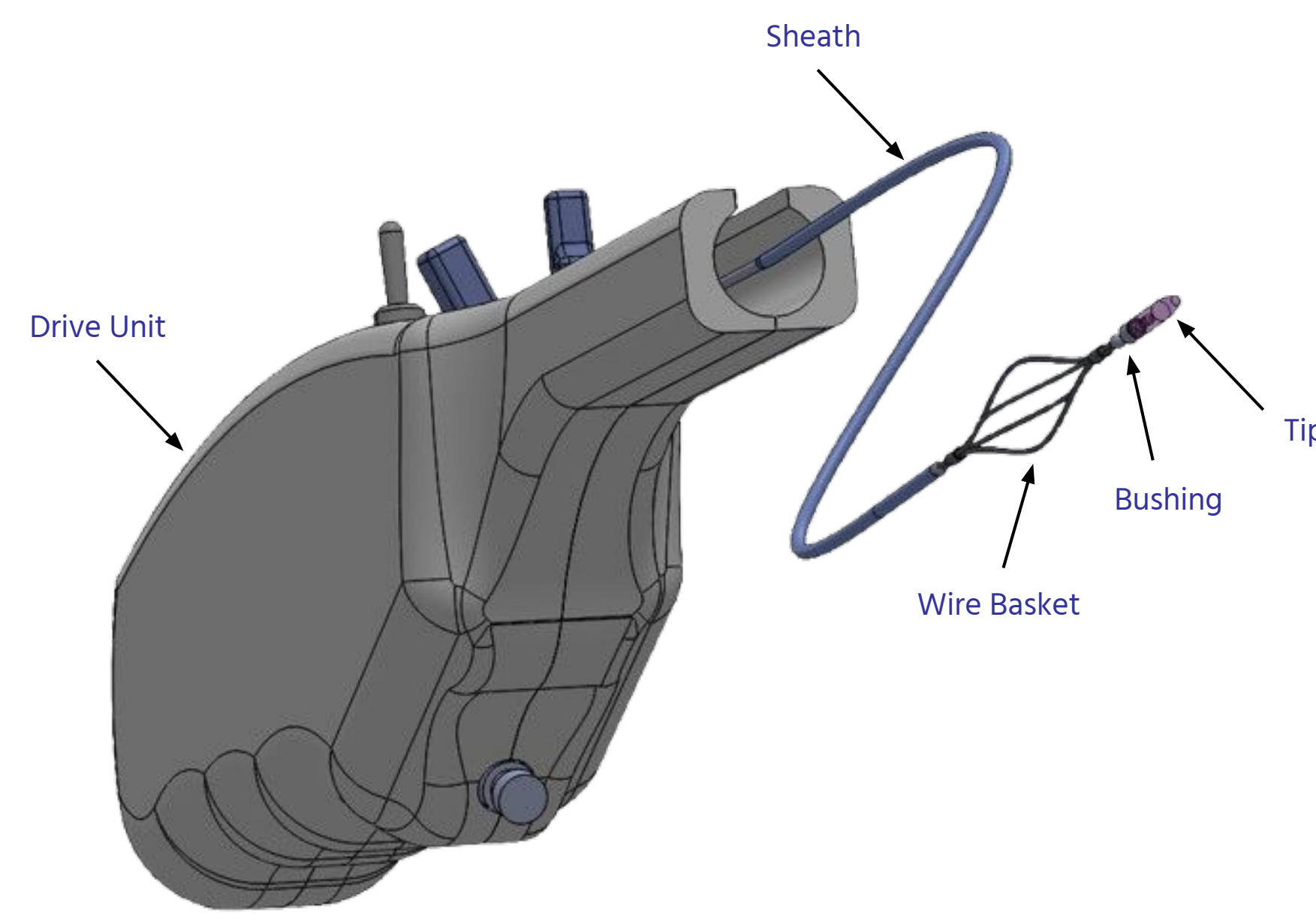


Fig. 9: CAD of Team's Prototype

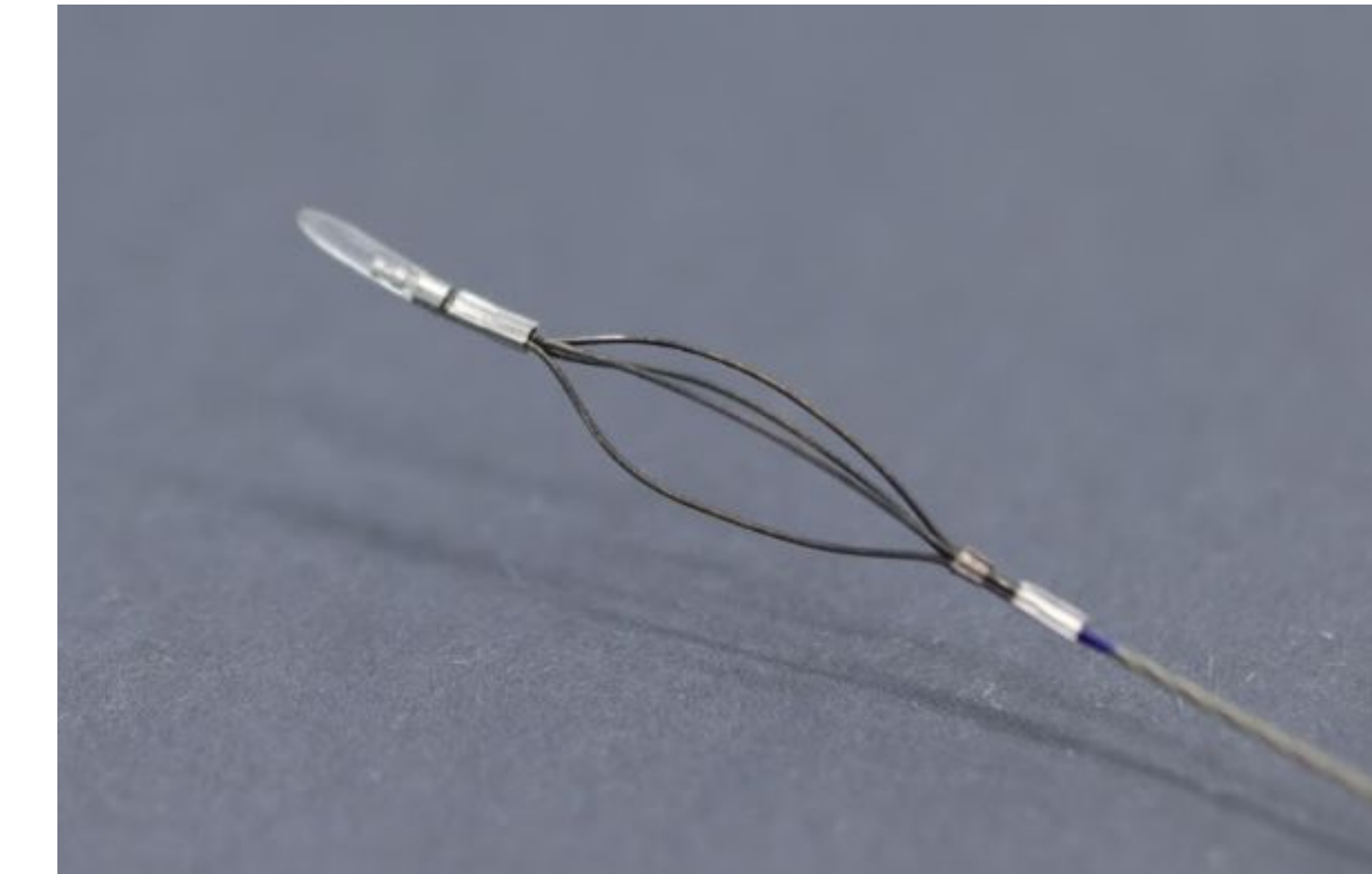


Fig. 10: Close-up photo of thrombolytic device

## Drive Unit

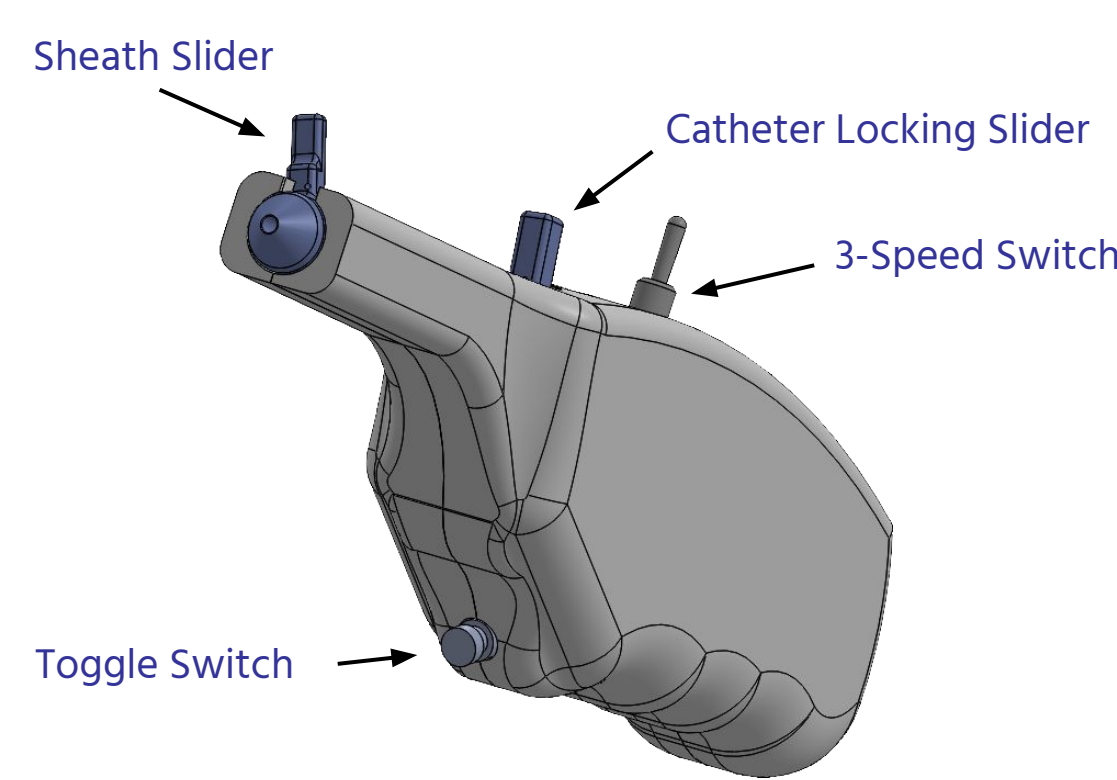


Fig. 10: Drive Unit CAD

- 3-speed control
- Ergonomic grip
- Interchangeable inserts for different basket designs

## Bushing

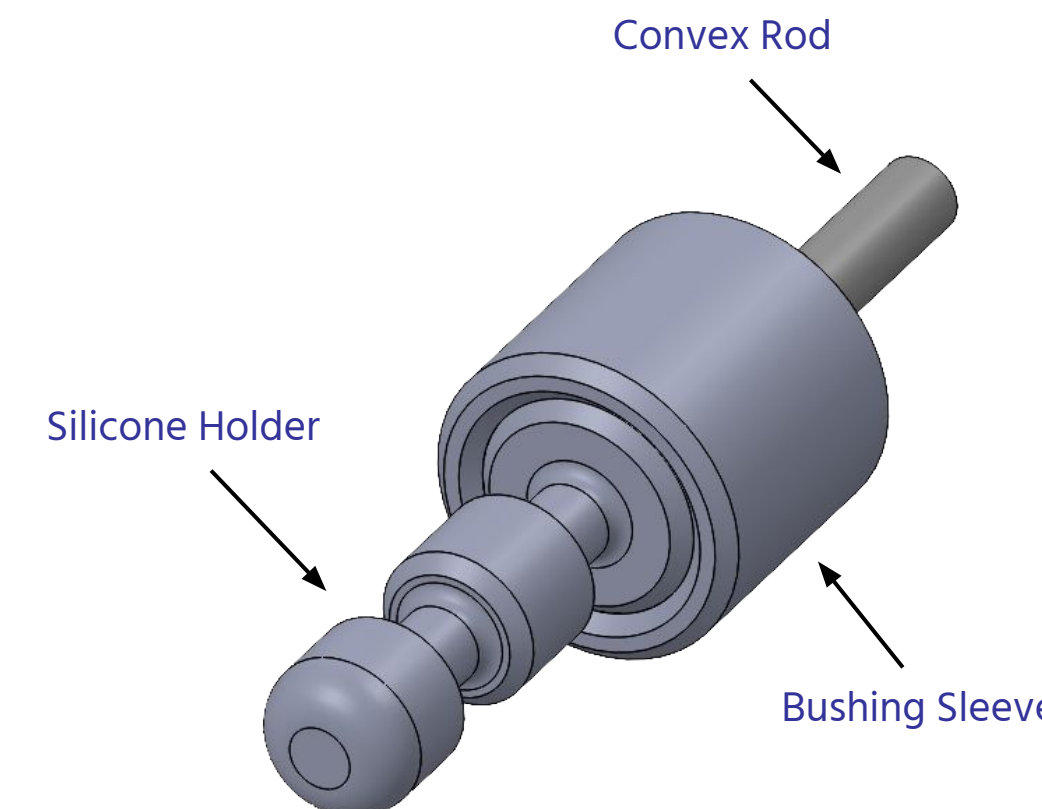


Fig. 11: Bushing Sub-assembly CAD

- Isolates basket rotation
- 2mm maximum outer diameter
- Custom geometry for DFM and tip connection strength

## Tip

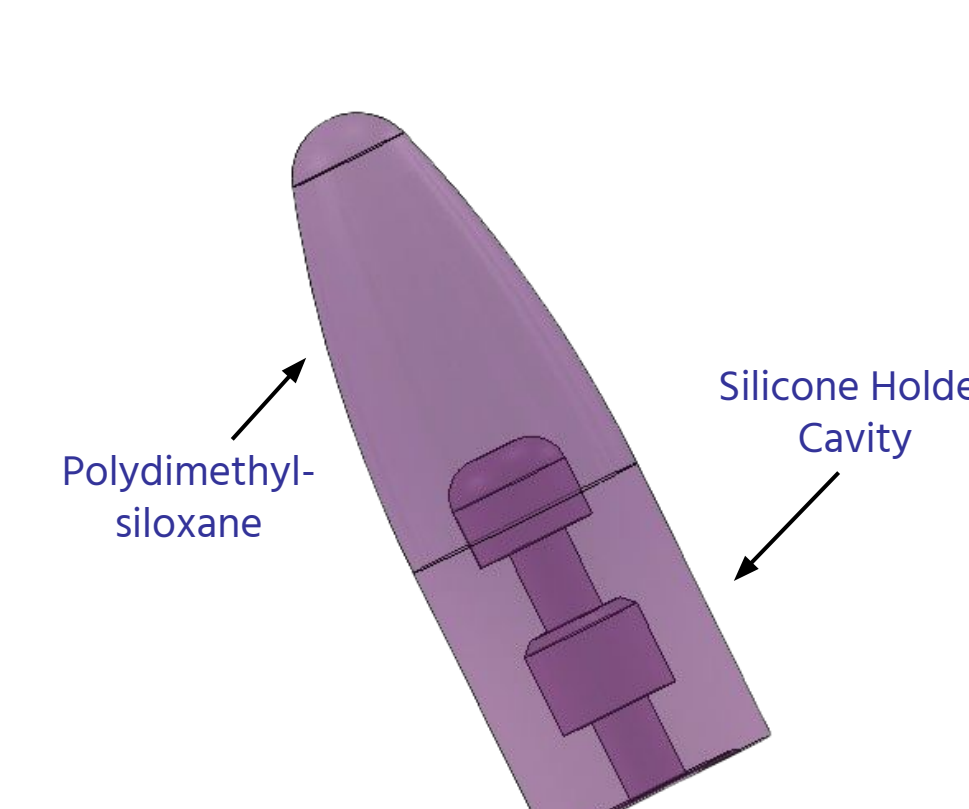


Fig. 12: Tip CAD

- Atraumatic to vessel wall
- Aids device maneuverability
- Enables device to puncture through clots

## Wire Basket

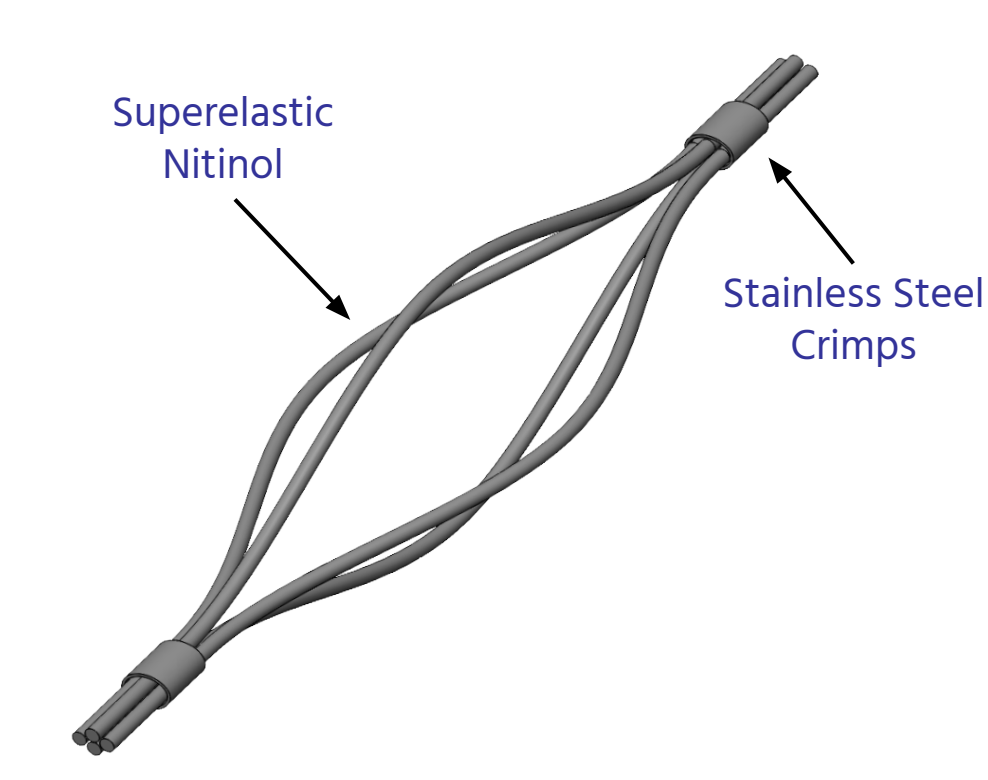


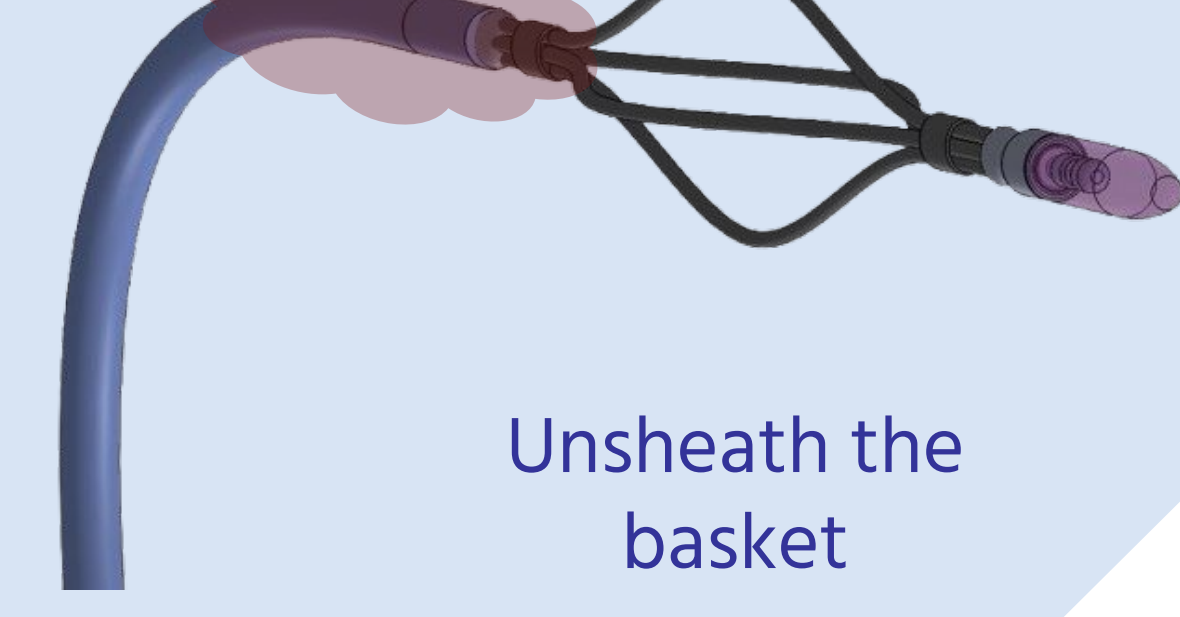
Fig. 13: Wire Basket Sub-assembly CAD

- Rapidly fragments clot
- Atraumatic to vessel at 5000rpm
- Clears clots in vessels of up to 8mm diameter

## Thrombectomy Procedure



### Step 2



### Step 3



## AV Graft Testbed

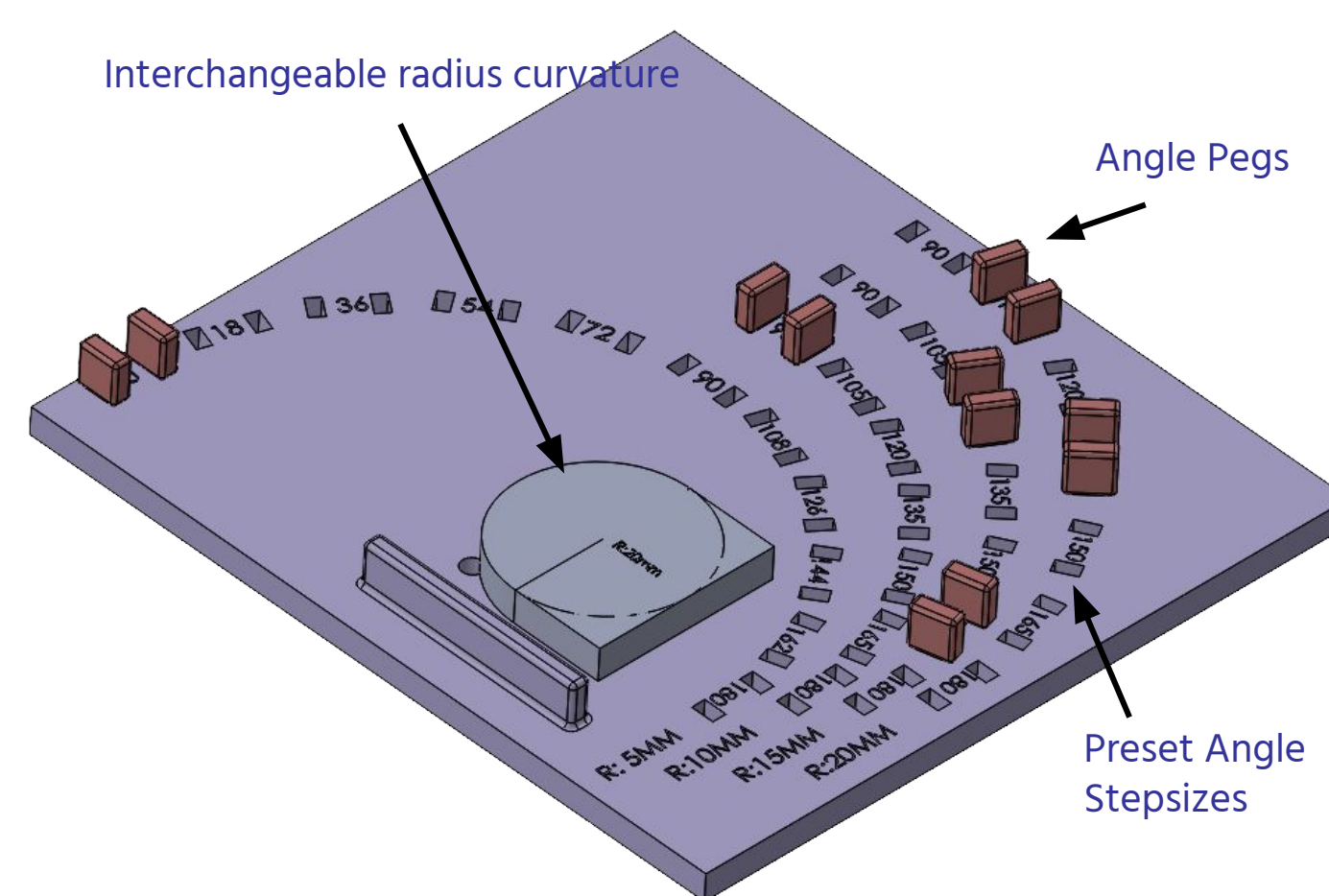


Fig. 14: CAD of Bend Testing Pegboard with various radii of curvature

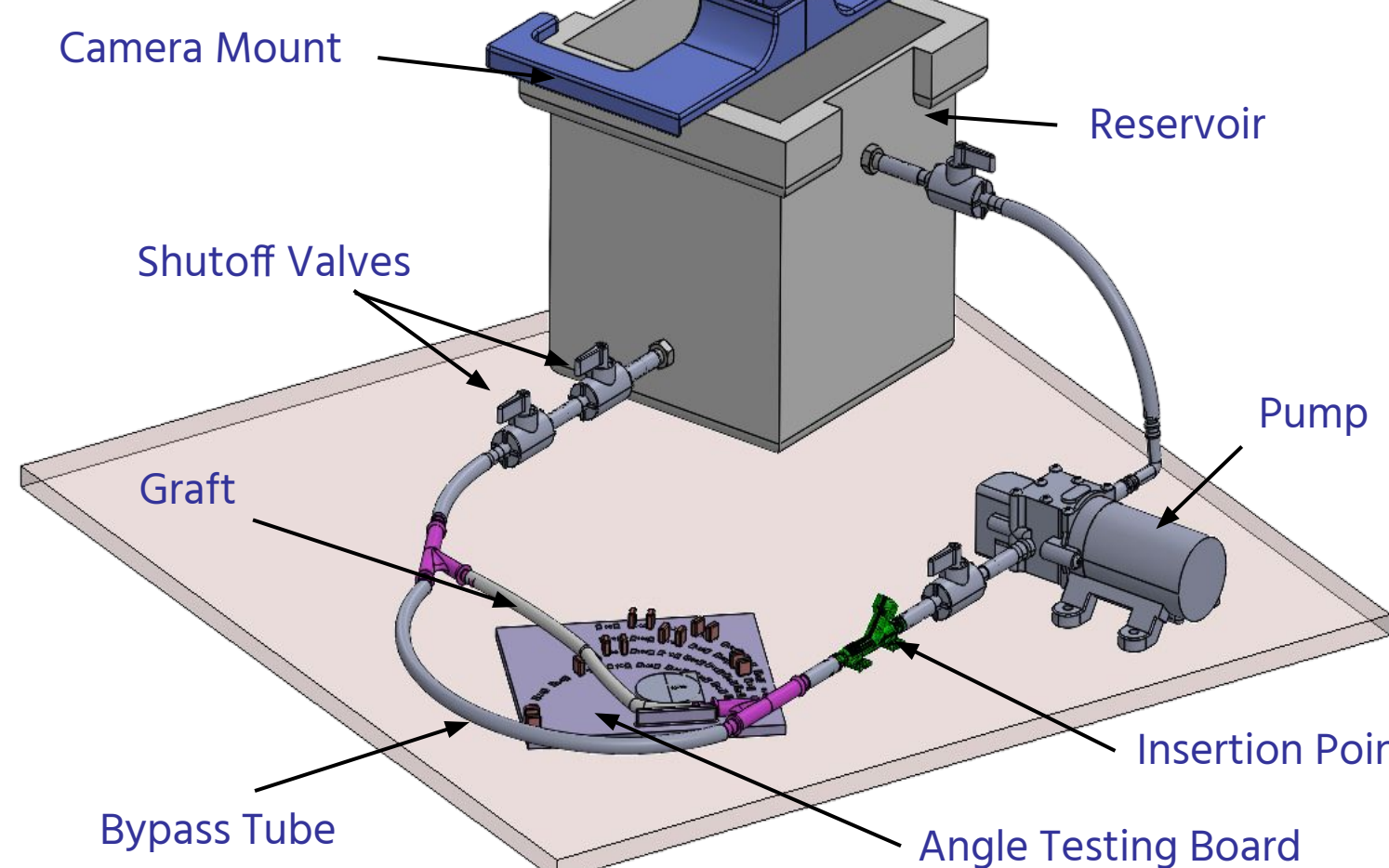


Fig. 15: CAD of Testbed

## Future Recommendations

- Improved material selection**
  - Source braided nitinol wire and marker bands
  - Customize silicone formula for tip
- Industrial manufacturing processes**
  - Replace press-fit with crimp
  - Hydraulic crimping press
- Optimization of bushing rotation**
  - Addition of silicone lubricant
  - Use of low friction polymer insert
- Iterative improvements**
  - Different baskets to target various clot compositions
  - Dedicated testbed module for CV software
  - Torque feedback on drive unit



Scan for more info about us!

## Test Results

|                      | % Clot Breakage under 2mm |
|----------------------|---------------------------|
| Cleaner              | 78.65%                    |
| Trerotola            | 82.64%                    |
| <b>Team's Design</b> | <b>79.45%</b>             |

Table 1: % Efficacy rates of devices tested

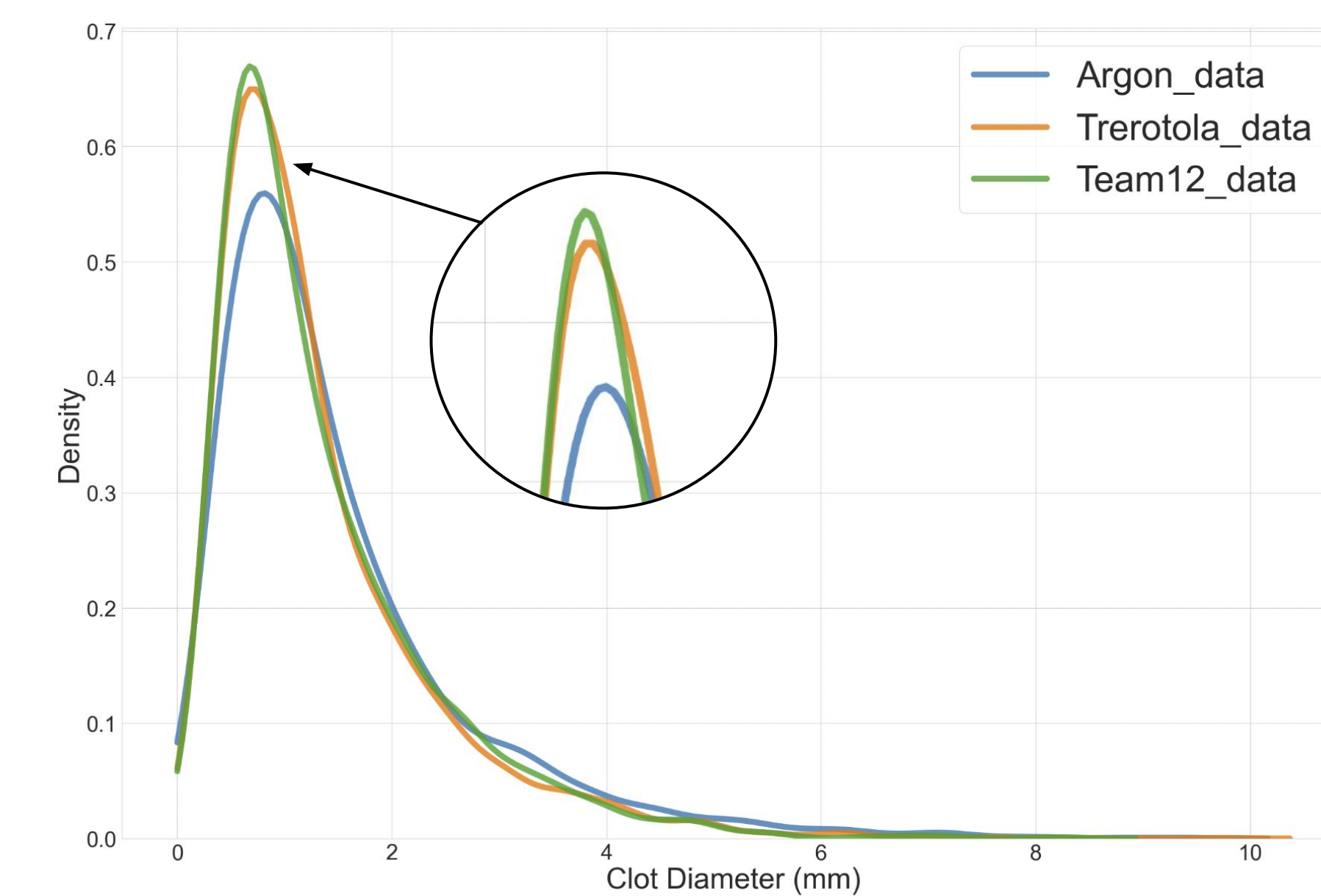


Fig. 16: Kernel Density Estimation of Broken Clot Size Distribution Between Different Devices



Fig. 17: Source photo and computer-vision analysis of circle contours over the fragmented clot of the team's design device

## Safety

- Mechanical removal:** fragments blood clots into pieces smaller than 2mm that can be naturally metabolized
- Biocompatible Material Selection:** Stainless Steel 304 & 316, Grade 2 Titanium, Superelastic NiTiNol, Sylgard 184
- FDA Regulations:** Class 2 Medical Device

## Impact on Society

- Provides **rapid clot removal**, reducing delays in resuming hemodialysis and associated complications
- Eliminates risk of tip breakage**, preventing device-related injuries and ensuring safer thrombectomy procedures

## Acknowledgements

The team would like to express appreciation to the following individuals for their support in making this project possible:

|                          |                 |                             |
|--------------------------|-----------------|-----------------------------|
| Prof. Nate Delson        | Dr. Elsie Ross  | Prof. Shengqiang Cai        |
| Prof. Jerry Tustaniwskyj | Kayla Tracy     | Prof. Ekaterina Evdokimenko |
| Jackelin Amorin          | Chris Cassidy   | Prof. Michael Frazier       |
| Jackie Chen              | Edward Pogue    | Prof. James Friend          |
| Prof. Maziar Ghazinejad  | Stephen Mercsak | Dr. Tamoghna Saha           |
| Prof. David Gillett      | Steven Roberts  | Dr. Omeed Djassemi          |
| Prof. Huihui Qi          | Tom Chalfant    | Carla Bacco                 |
| Ling Son                 | Aedan Mangan    | ... and many others         |

## References

- [1] NIDDK. "Kidney Disease Statistics for the United States." National Institute of Diabetes and Digestive and Kidney Diseases, 2023. [www.niddk.nih.gov/health-information/health-statistics/kidney-disease](https://www.niddk.nih.gov/health-information/health-statistics/kidney-disease).
- [2] Parisa Shiri, Shabab Rezaei, Alireza Abdi, Alireza Khatony, Prevalence of thrombosis in patients undergoing dialysis treatment: A systematic review and meta-analysis, Journal of Vascular Nursing, Volume 42, Issue 4, 2024, Pages 251-263, ISSN 1062-0303, <https://doi.org/10.1016/j.jvn.2024.08.003>.
- [3] Simpson, Russell O et al. "Managing Arrow-Trerotola Percutaneous Thrombolytic Device Complications." Seminars in interventional radiology vol. 36,2 (2019): 91-96. doi:10.1055/s-0039-1688421